

PURE POWERS 3^7 AND 3^8 IN THE INTEGER GROUP DETERMINANT IMAGE OF D_{54}

ABSTRACT. Let $\mathcal{S}(G)$ be the set of nonzero integer group determinants of a finite group G . We exhibit two elements $(x + x^4) + (x + x^4 - 1)s$ and $(1 + x + x^2 + x^4 + x^5) + (x + x^2 + x^4 + x^5)s$ of $\mathbb{Z}[D_{54}]$ whose group determinants are exactly $3^7 = 2187$ and $3^8 = 6561$, with no cofactor: pure powers of 3 with $v_2 = 0$. This is the new content of the note. Granting the necessity and achievability statements of Theorem 5.4 of Boerkoel and Pinner [1], which are quoted here and not reproved, these two pure powers close the exponent range $7 \leq b \leq 8$ left between those statements at $p = 3$, $k = 3$ and yield the description $\mathcal{S}(D_{54}) = \{2^a 3^b m : \gcd(m, 6) = 1, a = 0 \text{ or } a \geq 2, b = 0 \text{ or } b \geq 7\}$, m ranging over all of $\mathbb{Z}$.

1. THE WINDOW IN THEOREM 5.4

Let G be a finite group of order N and let $D_G(a) = \det(a_{uv^{-1}})_{u,v \in G}$ be the group determinant of $a = \sum_g a_g g \in \mathbb{Z}[G]$, an $N \times N$ integer determinant. Write

$$\mathcal{S}(G) = \{D_G(a) : a \in \mathbb{Z}[G]\} \setminus \{0\}$$

for the set of nonzero *measures* of G . Two conventions are inherited from [1]: the value $0 = D_G(0)$, attained for every G , is excluded throughout, and cofactors written m range over *all* of $\mathbb{Z}$ subject to the stated coprimality, not merely the positive integers. For the dihedral group

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$$

an element of $\mathbb{Z}[D_{2n}]$ is written $f(r) + g(r)s$ with $f, g \in \mathbb{Z}[x]/(x^n - 1)$.

Boerkoel and Pinner [1] proved the following.

Theorem 5.4. For $G = D_{2p^2}$ with $p = 3, 5$ or 7 the measures take the form $2^a p^b m$, $(m, 2p) = 1$ with $a = 0$ or $a \geq 2$, and $b = 0$ or $b \geq 5$.

In general, if $G = D_{2p^k}$ with p an odd prime and $k \geq 2$ the measures must take the form $2^a p^b m$, $(m, 2p) = 1$, with $a = 0$ or $a \geq 2$, and $b = 0$ or $b \geq 2k + 1$. We can achieve everything of this form with $b = 0$ or $b \geq 3k$. There are measures with $b = 2k + 1$.

Necessity is Lemma 4.4 of [1] (p. 9): if $p^\alpha \parallel n$ and $p \mid D$ then $p^{2\alpha+1} \mid D$. Achievability is assembled in the proof of Theorem 5.1 of [1] (pp. 15–16):

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Theorem 4.3 there gives every m coprime to $2p$, two displayed products give every 2^a with $a \geq 2$, and a family indexed by $\ell \geq k$ gives the pure powers $p^{\ell+2k}$, that is every p^b with $b \geq 3k$. That proof then remarks “For $k = 1$ these coincide and we have a complete description of the measures”, so the window

$$2k + 1 \leq b \leq 3k - 1$$

is not determined either way by Theorem 5.4 for every $k \geq 2$ (it lies strictly between the necessity bound and the achievability bound stated there). At $k = 2$ it is the single value $b = 5$, and the first clause of Theorem 5.4 covers it for $p = 3, 5, 7$. For $p = 3$ and $k = 3$, that is $G = D_{54}$ with $n = 27$, the window is $b \in \{7, 8\}$. Neither [1] nor its e-print poses a question or conjecture about this window; no priority or novelty claim is made here, and no literature search beyond [1] was carried out. What we prove is that 3^7 and 3^8 are measures of D_{54} with cofactor 1; the last sentence of Theorem 5.4 already asserts that some measure with $b = 2k + 1 = 7$ exists, and it is pure powers that the multiplicative assembly of [1] consumes.

2. THE TWO ELEMENTS

Throughout $n = 27$, ζ_d is a primitive d -th root of unity, $f^*(x) = f(x^{-1})$ in $\mathbb{Z}[x]/(x^{27} - 1)$, and

$$M(h) = \prod_{j=0}^{26} h(\zeta_{27}^j) = \text{Res}(x^{27} - 1, h)$$

is the circulant determinant of h . For n odd the reduction of [1] is

$$(1) \quad D_{D_{2n}}(f + gs) = M(ff^* - gg^*), \quad h := ff^* - gg^*.$$

Theorem 1. *Let $G = D_{54}$.*

$$(W1) \text{ For } f = x + x^4 \text{ and } g = f - 1 = -1 + x + x^4,$$

$$h = -1 + x + x^4 + x^{23} + x^{26}, \quad h(1) = 3,$$

$$\text{Res}(\Phi_3, h) = \text{Res}(\Phi_9, h) = \text{Res}(\Phi_{27}, h) = 9,$$

$$D_G(f + gs) = 3 \cdot 9^3 = 2187 = 3^7.$$

$$(W2) \text{ For } f = 1 + x + x^2 + x^4 + x^5 \text{ and } g = f - 1 = x + x^2 + x^4 + x^5,$$

$$h = 1 + (x + x^{26}) + (x^2 + x^{25}) + (x^4 + x^{23}) + (x^5 + x^{22}), \quad h(1) = 9,$$

$$\text{Res}(\Phi_3, h) = \text{Res}(\Phi_9, h) = \text{Res}(\Phi_{27}, h) = 9,$$

$$D_G(f + gs) = 9 \cdot 9^3 = 6561 = 3^8.$$

In both cases the value is a pure power of 3: $v_2 = 0$ and the cofactor after removing all factors 2 and 3 is 1.

3. THE VALUATION ARGUMENT

Both witnesses have the shape $g = f - 1$, and then $gg^* = (f - 1)(f^* - 1) = ff^* - f - f^* + 1$, so

$$(2) \quad h = f + f^* - 1, \quad \text{i.e.} \quad h_0 = 2a_0 - 1, \quad h_i = a_i + a_{27-i} \quad (1 \leq i \leq 26).$$

In particular h is symmetric, $h_i = h_{-i}$, and $h(1) = 2f(1) - 1$ is odd.

Lemma 2. *Let $h \in \mathbb{Z}[x]/(x^{27} - 1)$ be symmetric. Then for $m = 1, 2, 3$ there is $P_m \in \mathbb{Z}$ with $\text{Res}(\Phi_{3^m}, h) = P_m^2$, and*

$$(3) \quad v_3(D_{D_{54}}(f + gs)) = v_3(h(1)) + 2 \sum_{m=1}^3 v_3(P_m).$$

Moreover $3 \mid h(1)$ if and only if $3 \mid P_m$, for each m .

Proof. Symmetry gives $h(\zeta^{-1}) = h(\zeta) = \overline{h(\zeta)}$ for $\zeta = \zeta_{3^m}$, so $h(\zeta)$ lies in the maximal real subfield K_m^+ of $K_m = \mathbb{Q}(\zeta_{3^m})$, of index 2. Hence

$$\begin{aligned} \text{Res}(\Phi_{3^m}, h) &= \prod_{\zeta \text{ prim}} h(\zeta) = N_{K_m/\mathbb{Q}}(h(\zeta)) \\ &= \left(N_{K_m^+/\mathbb{Q}}(h(\zeta)) \right)^2 = P_m^2, \quad P_m \in \mathbb{Z}. \end{aligned}$$

Since $x^{27} - 1 = \prod_{m=0}^3 \Phi_{3^m}(x)$ and $\Phi_1(x) = x - 1$, equation (1) gives $D = h(1) P_1^2 P_2^2 P_3^2$, which is (3). Finally 3 is totally ramified in K_m , with $\pi_m = 1 - \zeta_{3^m}$ generating the prime above it and $N_{K_m/\mathbb{Q}}(\pi_m) = \Phi_{3^m}(1) = 3$; as $h(\zeta) \equiv h(1) \pmod{\pi_m}$, we get $\pi_m \mid h(\zeta)$ exactly when $3 \mid h(1)$, and $v_3(P_m^2) = v_{\pi_m}(h(\zeta))$. $\square$

Proof of Theorem 1. Both h displayed in Theorem 1 follow from (2) by inspection, and both have $h(1)$ divisible by 3; Lemma 2 then forces $v_3(P_m) \geq 1$ for $m = 1, 2, 3$. Since each $\text{Res}(\Phi_{3^m}, h) = 9$ we have $P_m = \pm 3$ and $v_3(P_m) = 1$, so (3) reads $v_3(D) = v_3(h(1)) + 6$, which is $1 + 6 = 7$ for W1 and $2 + 6 = 8$ for W2. The displayed products $3 \cdot 9^3 = 2187$ and $9 \cdot 9^3 = 6561$ are the values themselves, so nothing beyond 3 divides them and the cofactor is 1. $\square$

Remark (a hand check of P_1). Grouping the coefficients of h by residue mod 3 and using symmetry, $h(\zeta_3) = A - B$ with $A = \sum_{i \equiv 0} h_i$ and $B = \sum_{i \equiv 1} h_i = \sum_{i \equiv 2} h_i$, so $\text{Res}(\Phi_3, h) = (A - B)^2$ and

$$P_1 = h(1) - 3H_1, \quad H_1 = \sum_{\substack{1 \leq i \leq 13 \\ i \equiv \pm 1 \pmod{3}}} h_i.$$

For W1, $H_1 = h_1 + h_4 = 2$ and $P_1 = 3 - 6 = -3$; for W2, $H_1 = h_1 + h_2 + h_4 + h_5 = 4$ and $P_1 = 9 - 12 = -3$. Both give $P_1^2 = 9$.

Remark. Exchanging f and g replaces h by $-h$, and $M(-h) = -M(h)$ because 27 is odd; so -3^7 and -3^8 are measures as well.

4. COMPLETENESS

Theorem 3. *Granting the necessity and achievability statements of Theorem 5.4 of [1] at $k = 3$, $p = 3$, which are quoted and not reproved here,*

$$\mathcal{S}(D_{54}) = \{2^a 3^b m : m \in \mathbb{Z}, \gcd(m, 6) = 1, a = 0 \text{ or } a \geq 2, b = 0 \text{ or } b \geq 7\}.$$

Proof. The inclusion $\subseteq$ is Theorem 5.4 of [1] at $k = 3$: every measure is of the form $2^a 3^b m$ with $(m, 6) = 1$, $a = 0$ or $a \geq 2$, and $b = 0$ or $b \geq 2 \cdot 3 + 1 = 7$.

For $\supseteq$, D_G is multiplicative on $\mathbb{Z}[G]$, so $\mathcal{S}(G)$ is closed under multiplication. From [1] (proof of Theorem 5.1) $\mathcal{S}(D_{54})$ contains every $2^a m$ with $(m, 6) = 1$ and $a = 0$ or $a \geq 2$ — these are the $b = 0$ members — and every pure power 3^b with $b \geq 3k = 9$. Theorem 1 adds the pure powers 3^7 and 3^8 . The set of attainable exponents b therefore contains $\{0\} \cup \{7, 8\} \cup \{b : b \geq 9\} = \{0\} \cup \{b : b \geq 7\}$, and multiplying a pure power 3^b by a $b = 0$ member $2^a m$ realises every $2^a 3^b m$ of the stated form with $m > 0$. Finally, exchanging f and g replaces h by $-h$ and hence D by $-D$ (Remark at the end of Section 3), so $\mathcal{S}(D_{54})$ is closed under negation and the cases $m < 0$ follow from the cases $m > 0$. $\square$

Remark. Measures with $v_3 = 7$ and $v_3 = 8$ but nontrivial cofactor already follow from constructions printed in [1]; such cofactors cannot be removed afterwards, since $\mathcal{S}(G)$ is closed under multiplication and not under division.

5. REPRODUCTION

Everything above is determined by the two pairs (f, g) printed in Theorem 1 and by (1). To redo the computation: form $h = ff^* - gg^*$ in $\mathbb{Z}[x]/(x^{27} - 1)$ and evaluate $D = \text{Res}(x^{27} - 1, h)$ — equivalently, the 27×27 circulant determinant of h , or the product

$$h(1) \text{Res}(\Phi_3, h) \text{Res}(\Phi_9, h) \text{Res}(\Phi_{27}, h),$$

or the 54×54 determinant $\det(a_{uv-1})$ built directly from the multiplication table of D_{54} , which is the definition. All four routes agree on both witnesses.

The accompanying program `verify.py` carries out the witness computations in exact integer arithmetic: it builds D_{54} from its presentation, computes each determinant by all four routes, and confirms the valuations, the unit cofactors and the factorisations $P_m^2 = 9$. It does not verify the necessity statement or the infinite achievability results quoted from [1].

REFERENCES

- [1] M. Boerkoel and C. Pinner, *Minimal group determinants and the Lind–Lehmer problem for dihedral groups*, Acta Arith. **186** (2018), no. 4, 377–395. doi:10.4064/aa180708-30-8; e-print arXiv:1802.07336.